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Inverse design of dielectric metasurface transmitarray using physics-enhanced generative adversarial networks

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ABSTRACT

This paper proposes a physics-enhanced wasserstein generative adversarial network with gradient penalty for the inverse design of dielectric metasurface transmitarrays. Unlike existing approaches, the proposed network incorporates a differentiable simulator as a carrier of physical prior knowledge, achieving high spectral accuracy while preserving structural diversity. Furthermore, a lightweight physics regularization module is integrated to enforce sparsity and connectivity constraints, ensuring that the generated structures satisfy practical fabrication requirements. As a proof of concept, a dielectric metasurface transmitarray designed by the proposed network is fabricated using 3D printing technology and experimentally demonstrated for first-order orbital angular momentum beam generation. The measured results show good agreement with full-wave simulations, validating the feasibility and effectiveness of the proposed inverse design methodology.

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I. INTRODUCTION

The interaction between matter and electromagnetic waves underpins both natural optical phenomena and the design of novel devices.[1,2] Electromagnetic metamaterials, as artificial composites with unique properties,[3] offer new solutions to conventional antenna design bottlenecks. Integrating metamaterials into antennas enables miniaturization, bandwidth broadening, gain enhancement, and beam scanning.[4,5] However, traditional design methods, relying on empirical parameter sweeping or full-wave simulation, are computationally intensive and time-consuming, while failing to fully explore the design space.[6] Recently, deep learning has advanced rapidly, showing great promise in image classification,[7] computer vision,[8] and natural language processing.[9] Meanwhile, many researchers have applied deep learning to metasurface optimization.[10] Recent works have also explored programmable metasurfaces for sensing and communication applications, such as robust spatiotemporal 3D human pose estimation,[11] and integrated sensing and communication based on space-time-coding metasurfaces.[12]

Compared to forward spectral prediction, the inverse design of metasurfaces is more challenging because of its ill-posed nature. Specifically, the one-to-many mapping from target spectra to geometric structures complicates the problem.[13] In addition, the complex resonance behavior within the operating band hinders neural networks from accurately establishing the inverse geometric–electromagnetic mapping. Consequently, phase deviations from the targets are often observed. As a generative model, generative adversarial networks (GANs) show promise in metasurface inverse design by learning data distributions and generating structures.[14,15] Liu *et al.*[16] proposed an unsupervised method integrating contrast source inversion with GAN, eliminating the need for labeled datasets and enabling adaptive, efficient data acquisition.

Building on this, we propose a physics-enhanced wasserstein generative adversarial network with gradient penalty (PE-WGAN-GP) for metasurface inverse design. A differentiable

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simulator is introduced as physical prior, shifting the generator's objective from mimicking training data to producing designs that yield target spectral responses, thereby ensuring spectral accuracy while enabling structural diversity. A lightweight physics regularization module enforces manufacturability via sparsity and connectivity constraints. As a proof of concept, we design and fabricate a dielectric metasurface transmitarray for first-order vortex wave generation using the proposed network. Experimental results agree well with full-wave simulations, confirming that the generated design accurately achieves the target phase distribution. This work provides an intelligent and efficient optimization approach for metasurface unit, with potential value in novel functional device development.

II. METASURFACE UNIT DESIGN

Figure 1 illustrates the proposed unit structure of the metasurface. The dielectric unit is fabricated using polylactic acid (PLA) material, which exhibits a dielectric constant of 2.72 and a loss tangent of 0.015. The unit has a uniform height of $H = 25$ mm and a period of $P = 12$ mm. Limited by the fabrication precision of the current 3D printing equipment, the dielectric unit is discretized perpendicular to the z axis into a binary image, represented by a 20×20 Boolean matrix, where "1" denotes the dielectric material and "0" denotes air. Additionally, the unit is designed to exhibit quarter-mirror symmetry with respect to its top-left quadrant. To ensure manufacturability, the unit structure is designed to be fully connected, with the additional constraint that the outermost perimeter consists entirely of the dielectric material, thereby enabling the array to form an integrated monolithic structure.

III. DATA GENERATION AND NETWORK DESIGN

A. Data acquisition

Dielectric unit typically achieves phase modulation by adjusting the equivalent height or equivalent thickness of the dielectric material. Therefore, considering the precision constraints of 3D printing, the fill ratio is adopted as one of the criteria to generate the unique matrix corresponding to the unit topology. Additionally, breadth-first search (BFS) is employed to ensure the connectivity of the generated structures. To cover as many fill-ratio cases as possible with a limited number of samples, the complementary structure of each unit is also generated, with the connectivity of the complementary unit ensured. Subsequently, co-simulation was performed by integrating MATLAB with CST Microwave Studio to obtain the spectral response curves corresponding to the dielectric unit. During the simulation process, floquet periodic boundary conditions were applied in the x and y directions, while open boundary conditions were imposed in the z direction. The pseudocode of the algorithm for co-simulation dataset generation based on the BFS algorithm is shown below, and the overall flow chart is illustrated in Fig. 2.

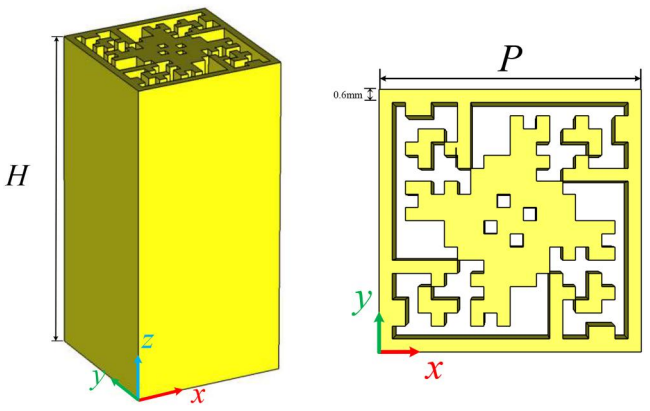


FIG. 1. The proposed structure and parameters of the dielectric metasurface unit.

B. Network design and training

The network architecture employed in this study is a physics-enhanced generative adversarial network. As illustrated in Fig. 3, the network framework consists of three core components: a generator, a discriminator, and a differentiable simulator, integrated with physics-based regularization constraints, enabling end-to-end mapping from target spectra to structural designs.

Among them, the generator adopts a conditional generative architecture, taking a target spectrum vector and a random noise vector as inputs and outputting a binary design matrix. The input spectrum is mapped into a latent feature representation through three fully connected layers, with the feature dimensionality

Algorithm: Data Collection for Symmetric Connected MSA

Input : Total number of samples: N; grid size: $N_x \times N_y$

Output: Dataset containing topology, response curve

for $n = 1$ *to* N **do**
  Initialize UL as an $N_x \times N_y$ zero matrix;
  Set $UL(:,1) = 1$ and $UL(1,:) = 1$;
  for $i = 2$ *to* N_x **do**
    for $j = 2$ *to* N_y **do**
      $UL(i,j) = 1$ with probability random$(0.1, 0.9)$, else 0;
    end
  end
  Initialize *visited* as an $N_x \times N_y$ zero matrix, $queue = \emptyset$;
  $visited(1,1) = 1$, Enqueue $(1,1)$;
  while *queue is not empty* **do**
    (row, col) = Dequeue;
    foreach (dr, dc) *in* $\{(-1,0),(1,0),(0,-1),(0,1)\}$ **do**
      $r = row + dr, c = col + dc$;
      if $1 \le r \le N_x$ *and* $1 \le c \le N_y$ *and* $UL(r,c) == 1$ *and* $visited(r,c) == 0$ **then**
        $visited(r,c) = 1$, Enqueue (r,c);
      end
    end
  end
  $UL_connected = UL \cdot visited$;
  $final_matrix = \begin{bmatrix} UL_connected & \text{rot90}(UL_connected, 3) \\ \text{rot90}(UL_connected) & \text{rot90}(UL_connected, 2) \end{bmatrix}$;
  Invoke co-simulation, simulate response curve, store into dataset;
end

FIG. 2. Co-simulation dataset generation based on the BFS algorithm.

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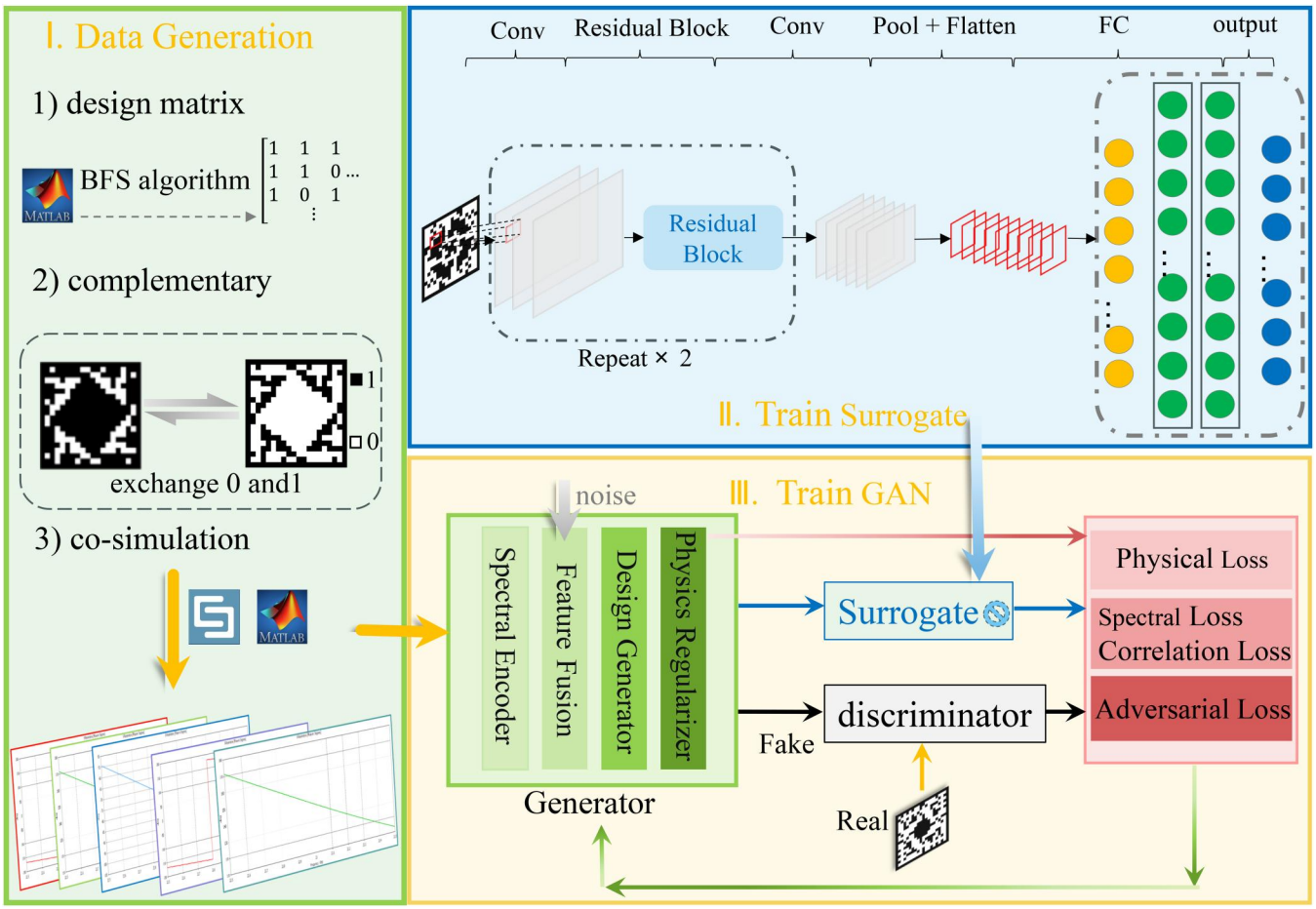


FIG. 3. Workflow of the GAN-based inverse design for metasurfaces.

progressively decreasing layer by layer. The encoded spectral features are then concatenated with the random noise along the feature dimension, and finally the fused features are mapped to the design matrix. The discriminator network adopts the wasserstein GAN with gradient penalty (WGAN-GP) architecture. The network consists of a convolutional feature extractor and a fully connected classifier, outputting a discrimination score to guide the adversarial training of the generator. To mitigate the non-uniqueness issue in metasurface inverse design, the simulator is designed to shift the optimization objective. Specifically, instead of "generating designs similar to those in the training set," the objective becomes "generating designs that yield the target spectral response." In this way, spectral accuracy is ensured while diverse representations of the designed structures are allowed. To enhance feature extraction capability and alleviate the vanishing gradient problem, residual blocks are embedded to facilitate information flow while maintaining feature dimensions. The simulator is first pre-trained on pairs of real data to learn an accurate physical mapping.

During the simulator pre-training phase, this study designs a phase-aware loss function, as given in Eq. (1), which consists of two components: an amplitude loss and a gradient loss. The amplitude loss constrains the absolute numerical error between the predicted spectrum and the target spectrum. The gradient loss, on the other hand, constrains the rate of change of the spectrum, which corresponds to the physical group delay characteristics, thereby ensuring the smoothness and continuity of the spectrum. The gradient loss weight is set to $\lambda = 0.3$, which was empirically determined through preliminary experiments: a smaller λ tends to produce spectra with noticeable local oscillations, whereas a larger λ over-smooths the spectrum and degrades the amplitude accuracy. The value $\lambda = 0.3$ provides a good balance between amplitude fidelity and spectral smoothness, and has been adopted throughout this work,

$$L = \|\hat{s} - s_{\text{target}}\|_2^2 + \lambda \|\nabla \hat{s} - \nabla s_{\text{target}}\|_2^2. \quad (1)$$

The simulator is pre-trained on the same dataset to learn the forward mapping from structural designs to spectral responses.

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Specifically, the simulator is optimized using the Adam optimizer with an initial learning rate of 5×10^{-5}, betas of (0.5, 0.9), and a weight decay of 1×10^{-5}. The batch size is set to 8. The model is trained for up to 150 epochs with an early-stopping patience of 20 based on the validation loss. To verify the fidelity of the pre-trained simulator, we evaluated its standalone prediction accuracy on the held-out test set. For each test sample, the real structural design was fed into the simulator, and the predicted spectrum was compared with the ground-truth spectrum. The simulator achieves a median correlation coefficient of 0.9984 and a first quartile of 0.9913, indicating that the majority of predictions are highly accurate. These results confirm that the pre-trained simulator provides reliable physical constraint signals for the generator.

After pre-training, the simulator parameters are fixed to provide differentiable physical constraint signals for the generator. To ensure the physical realizability of the generated designs, a lightweight physics-based regularization module is integrated into the network. During training, this module primarily enforces connectivity of the structures by constraining the corner and edge regions of the design matrix. Combined with post-processing validation, this approach ensures both inter-unit and intra-unit connectivity while offering high computational efficiency and differentiability.

The three core networks achieve inverse mapping from spectrum to design through joint optimization. The training process adopts a three-stage strategy: (1) the simulator is first pre-trained to learn an accurate physical mapping; (2) adversarial training is then performed within the GAN framework, where the generator learns to produce physical designs that can both deceive the discriminator and be validated by the simulator; (3) finally, a physics-based regularization module ensures the manufacturability of the generated designs. The overall network framework achieves an effective integration of physical knowledge, adversarial learning, and end-to-end optimization, providing an efficient and reliable solution for metasurface design.

To comprehensively evaluate the generative performance of the proposed model, a multi-metric evaluation system is constructed from two perspectives: spectral matching quality and physical structural rationality. In terms of spectral matching, the correlation coefficient is employed to quantify the shape similarity between the generated spectrum and the target spectrum. It is defined as the Pearson correlation coefficient between the predicted spectrum $\hat{s}_i$ and the target spectrum s_i, as shown in Eq. (2). where B is the batch size and F is the number of frequency points. The phase error directly reflects the model's prediction deviation in physical phase and is computed as Eq. (3). The intercept error assesses the phase prediction accuracy at the initial frequency point, as shown in Eq. (4). In addition to global metrics, the intercept error is introduced as a local diagnostic indicator to evaluate the accuracy of phase prediction at the initial frequency points. Regarding physical rationality, internal connectivity is ensured through the analysis of unit connected domains, and adjacent units' connectivity is ensured by determining corners and edges, thereby ensuring that the entire metasurface unit can be manufactured, with connectivity loss used to measure the degree of connectivity of the units. Figure 4 illustrates the evolution curves of the aforementioned evaluation metrics throughout the training process. It can be observed that as the number of training epochs increases, the spectral error gradually decreases and eventually converges. The connectivity loss ultimately reaches zero, indicating that the network is capable of generating fully connected units,

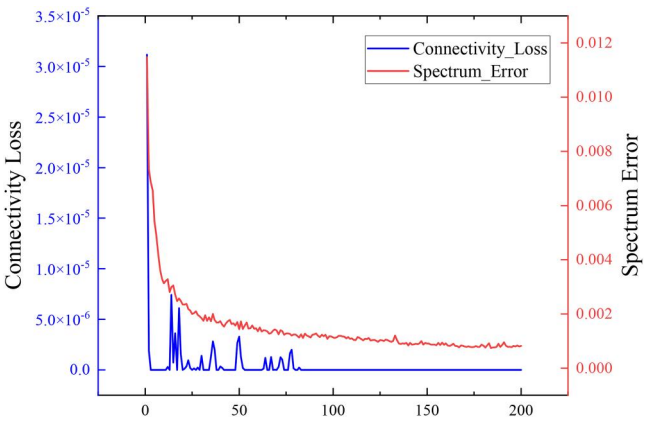


FIG. 4. Curves of various monitoring metrics during the network training process.

$$\rho = \frac{1}{B}\sum_{i=1}^{B} \frac{\sum_{j=1}^{F}\left(\hat{s}_i(j)-\bar{\hat{s}}_i\right)\left(s_i(j)-\bar{s}_i\right)}{\sqrt{\sum_{j=1}^{F}\left(\hat{s}_i(j)-\bar{\hat{s}}_i\right)^2}\sqrt{\sum_{j=1}^{F}\left(s_i(j)-\bar{s}_i\right)^2}}, \tag{2}$$

$$\mathscr{L}_{\text{phase}} = \frac{1}{B\cdot F}\sum_{i=1}^{B}\sum_{j=1}^{F}\left|\phi_i^{\text{pred}}(j)-\phi_i^{\text{target}}(j)\right|, \tag{3}$$

$$\mathscr{L}_{\text{intercept}} = \frac{1}{B}\sum_{i=1}^{B}\left|\phi_i^{\text{pred}}(1)-\phi_i^{\text{target}}(1)\right|. \tag{4}$$

To promote stable convergence of model training, this paper adopts a cosine annealing strategy for dynamic learning rate scheduling, which enables the learning rate to decay smoothly along a cosine curve, thereby avoiding the loss oscillation issues that may arise from conventional stepwise decay strategies. Furthermore, a training-state-aware dynamic adversarial weight adjustment mechanism is designed, which adaptively tunes the adversarial loss weight without requiring manual intervention. Specifically, when the improvement in phase error falls below 5% over 20 consecutive training epochs, the system automatically amplifies the adversarial loss weight by a factor of 1.2. By enhancing the gradient signal from the discriminator, this mechanism compels the generator to overcome the current optimization bottleneck, consequently improving the spectral matching capability and physical rationality of the generated structures.

Figures 5 and 6 present two metasurface units designed using the proposed network model. These units were simulated in electromagnetic simulation software, and the obtained actual responses were compared with the intended responses. It can be

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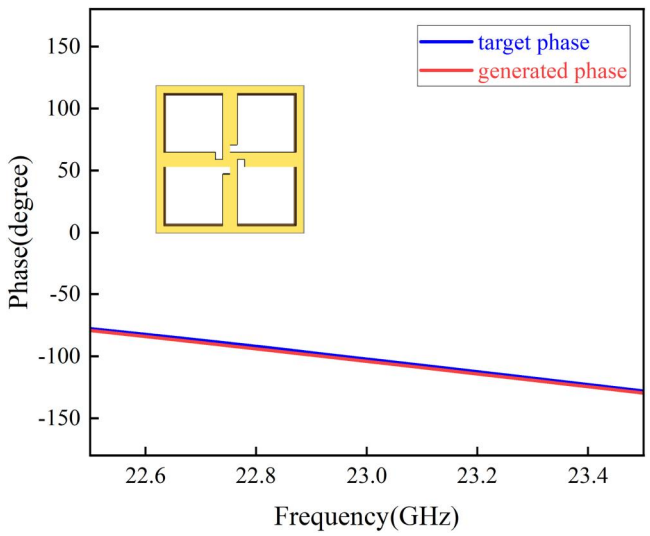


FIG. 5. Comparison of target and generated phases for metasurface unit using the proposed network.

observed that for Fig. 6, the target phase curve and the actual phase curve exhibit a high degree of shape consistency. For Fig. 5, the target phase and the actual phase are nearly coincident throughout. The difference in results between the two phase intervals is mainly attributed to whether the target phase falls in a dense or sparse region of the dataset. Figure 7 shows the final generated metasurface unit cell for the array. Table I provides a comprehensive comparison of our work with other similar studies.

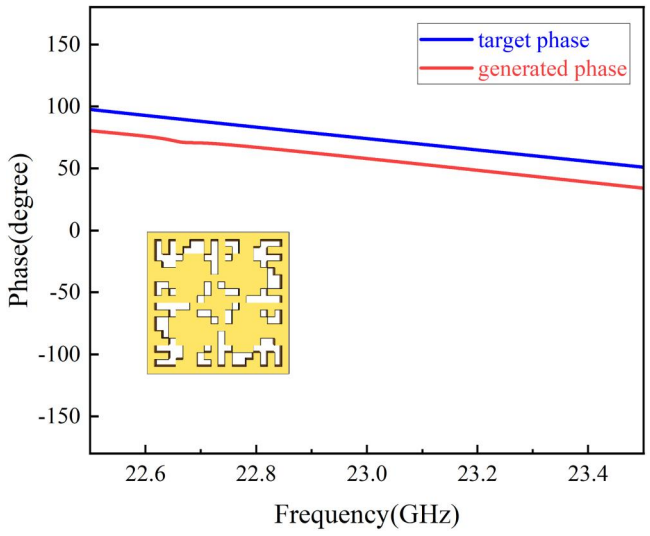


FIG. 6. Comparison of target and generated phases for metasurface unit using the proposed network.

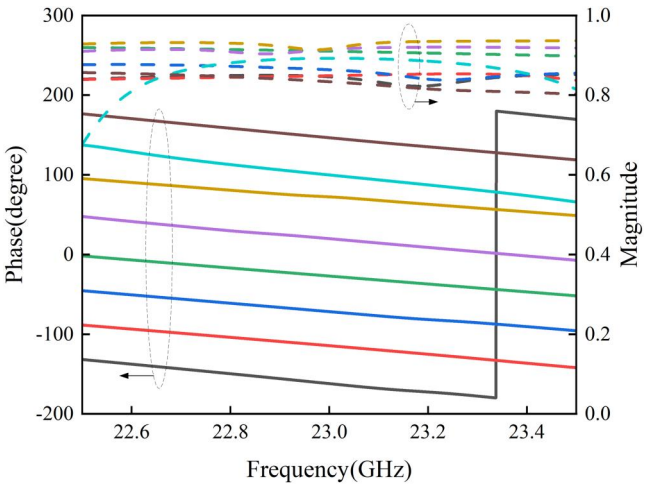


FIG. 7. The metasurface unit ultimately generated for constructing the array.

The dataset was randomly shuffled and then divided into a training set and a test set at a ratio of 8:2. To evaluate the generalization capability of the proposed model. The test set was strictly held out and was not used during training or hyperparameter tuning. To investigate the generalization performance across different phase intervals, we divided the test set into six intervals of 60° each. The results show that the model maintains high accuracy in well-sampled intervals, a correlation coefficient of 0.9969 in [−120°, −60°) and 0.9968 in [−180°, −120°), while the performance slightly decreases in sparsely sampled intervals, e.g., 0.9845 in [60°, 120°). This further confirms that the proposed model generalizes well across the sampled phase range, and that the residual performance imbalance is mainly attributed to the non-uniform distribution of the training data rather than overfitting.

IV. OAM-GENERATING TRANSMITARRAY SIMULATION AND MEASUREMENT

Orbital angular momentum (OAM) vortex waves, as a novel transmission carrier, open up a new modulation dimension for electromagnetic spectrum multiplexing. This technology not only effectively improves spectral efficiency in wireless communications but also demonstrates unique advantages in scenarios such as radar detection and target recognition.

TABLE I. Comparison with other neural network-based metasurface inverse design methods.

Reference	Arch.	Samples	Phys. Cons.	Acc.
17	WGAN-GP	30k	None	MSE 0.03
18	CGAN	20k	None	96.64%
19	DNN+SDAL	2.6k	None	90.84%
This work	WGAN-GP +Surr.	3.6k	Spars. +Conn.	0.9921

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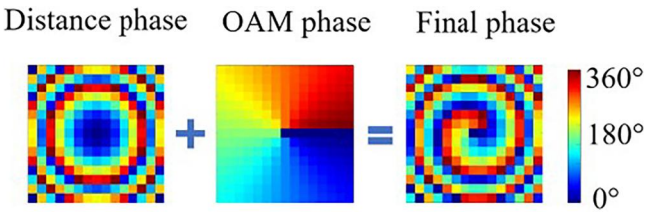


FIG. 8. The phase synthesis of the OAM vortex waves.

Based on the above modeling approach, metasurface unit with desired phase responses can be obtained, which are then employed to construct an OAM-generating transmitarray. This section describes the design procedure for a first-order OAM-generating transmitarray. As illustrated in Fig. 8, to compensate for the path phase differences arising from distinct spatial locations, the final compensation phase is determined by superposing the spatial distance phase and the OAM helical phase.

Following the initial numerical validation, the transmitarray prototype was manufactured via 3D printing. A Bambu P1S printer was utilized, and the dielectric material was selected as PLA. To ensure the fidelity of the printing process, the 15×15 transmitarray was divided into four parts, corresponding to the upper-left, upper-right, lower-left, and lower-right quadrants. The resulting physical prototype is presented in Fig. 9.

Measurements of the antenna were performed in a microwave anechoic chamber, as illustrated in Fig. 9. The measurement setup comprised a waveguide probe, the transmitarray, and a horn antenna, with the centers of all three components aligned collinearly. The probe-to-transmitarray distance was 450 mm, while the feed horn was positioned 160 mm away from the transmitarray. Figure 10 shows the simulated phase and far-field 3D radiation patterns of the first-order vortex beam over 22.5–23.5 GHz, together with the measured phase and amplitude results. The measured amplitude distribution is relatively flat, with a near-zero field at the beam center, forming a characteristic doughnut-shaped profile. The measured phase distribution exhibits the typical helical phase front of a vortex beam. Overall, the experimental results are found to be in excellent agreement with the numerical simulations.

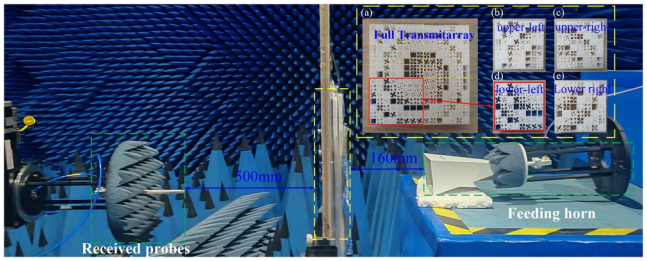


FIG. 9. Measurement photograph of the 3D-printed transmitarray in an anechoic chamber.

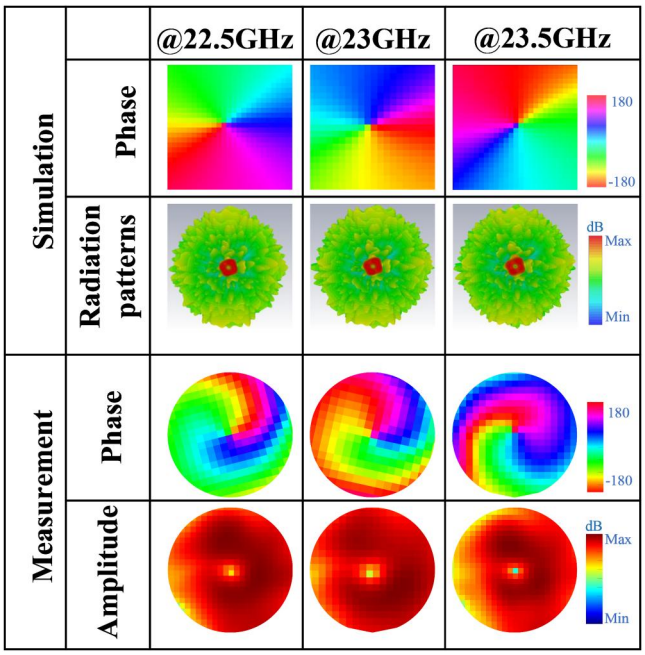


FIG. 10. Photograph of the 3D-printed transmitarray under test in an anechoic chamber.

V. CONCLUSION

In this study, an end-to-end inverse design method from target spectra to metasurface structures is achieved based on the WGAN-GP framework integrating physical constraints and a surrogate simulator, attaining an overall correlation coefficient of 0.9921. Using this network, a transmitarray capable of generating first-order vortex beams is further designed and fabricated, and the simulated results are in basic agreement with measurements, validating the feasibility of the network in practical engineering.

Nevertheless, the generation performance of the network remains unbalanced across different phase intervals. Statistical analysis of the dataset reveals a markedly non-uniform phase distribution: The normalized distribution entropy is 0.836 (1.0 for a perfectly uniform distribution). In contrast, the structural matrices are nearly uniformly distributed, indicating that the non-uniformity originates from the target phase distribution rather than the structural data. Due to the limitations of 3D printing accuracy, when designing units, the minimum size is best kept >=0.6 mm (the length corresponding to each small grid in the units used in this study is 0.6 mm), which limits the achievable phase range. Most phase samples in the training set are concentrated in [−120° to 0°], and some intervals have fewer samples, causing insufficient training in sparse intervals, ultimately leading to uneven generation performance across different phase ranges. Future work should address this issue through strategies such as active learning and data augmentation.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Qi Liu: Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Validation (equal); Writing – original draft (equal); Writing – review & editing (equal). **Gaonian Zhang:** Formal analysis (equal). **Yue Ma:** Formal analysis (equal). **Jiang Yang:** Methodology (equal). **Zhixiang Huang:** Funding acquisition (equal). **Guanghui Xu:** Conceptualization (equal); Funding acquisition (equal); Methodology (equal). **Menglin Zhai:** Funding acquisition (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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